Chuwen Xu

Undergraduate Student - School of Mathematics and Statistics - Wuhan University

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Formal Education

Graduation in progress in Mathematics

School of Mathematics and Statistics - Wuhan University (WHU)

September 2024 – Ongoing | Wuhan, China

- **GPA:** 3.92 / 4.0
- **Selected Coursework:**
  Mathematical Analysis (**97**), Advanced Algebra (**97**), Ordinary Differential Equations (**97**), Abstract Algebra (**93**), Complex Variables (**93**), Probability Theory (**92**), General Physics (**94**), Data Structures & Algorithmic Design in Python (**99**), Mathematical Modeling (**93**), Biopharmaceutical Technology (**92**).

Complementary Education

Academic Exchange and Research Practice Program

The Chinese University of Hong Kong (CUHK)

March 2026 | Hong Kong, China

- Attended advanced academic seminars presented by Prof. Jun Zou and Prof. Ronald Lok Ming Lui, focusing on **quasiconformal mapping**, **inverse scattering problems**, and their engineering applications.
- Engaged in intense inter-university academic forums, exploring global Ph.D. training modes, research methodologies, and potential future doctoral dissertation topics with CUHK faculty and graduate students.

Cross-disciplinary Applied Mathematics Symposium

The Hong Kong Polytechnic University (PolyU)

March 2026 | Hong Kong, China

- Investigated the cutting-edge applications of applied mathematics in cross-disciplinary frontiers, including **numerical solutions of PDEs**, **quantitative finance**, and **deep learning-driven material science**.
- Conducted face-to-face academic networking with prominent professors and alumni regarding advanced research methodologies, graduate admissions requirements, and global fellowship applications.

Advanced Summer School on Large Language Models and Generative Learning

Southwestern University of Finance and Economics (SWUFE)

July 2026 | Chengdu, Sichuan, China

- Attended a series of intensive graduate-level short courses on deep learning theory, LLMs, and modern generative models delivered by prominent scholars from **UCSB, UCLA, UC Irvine, PolyU**, and **Tsinghua University**.
- Explored foundational deep learning theory, including mathematical function approximation via DNNs, continuous normalizing flows, and the connections between statistical mechanics and deep models.
- Studied advanced reinforcement learning alignment frameworks (RLHF/DPO), optimal control theory in diffusion model fine-tuning, and algorithmic robustness/privacy under adversarial attacks.
- Investigated cross-disciplinary AI deployment methodologies, specifically focusing on Multi-Agent LLMs planning, Retrieval-Augmented Generation (RAG), and domain-specific applications in sports analytics.

Areas of Interest

- **Computational Biology:** Focused on developing robust statistical frameworks and scalable alignment algorithms applied to high-throughput sequencing data across transcriptomics, genomics, and proteomics.
- **Machine Learning & Deep Learning Theory:** Interested in probabilistic graphical models, high-dimensional statistical inference, and deep learning-based clustering to unravel complex mechanisms in heterogeneous biological systems.
- **Large-scale Data Mining:** Applied to unsupervised representation learning, manifold learning, and network/graph-based fusion for the integration and dimensionality reduction of massive, sparse multi-omics datasets.

Honors & Awards

National Scholarship

Ministry of Education

National | 2025

- The most prestigious nationwide honor awarded to the **top 0.2%** of students for exceptional academic excellence.

First-Class Merit Scholarship

Wuhan University

Institutional | 2025

- Awarded alongside the **"Three-Good Student"** title (**Top 5%**) for concurrent excellence in character.

Languages & Test Scores

- **English** — **CET-6: 569** (4/5)
- **Mandarin Chinese** — **Native Speaker** (5/5)

Skills & Tools

- **Python** — **Data Analysis / ML** (5/5)
- **R Language** — **Bioinformatics / Stats** (4/5)
- **LaTeX** — **Academic Typesetting** (5/5)
- **Linux / Bash** — **Server Management** (4/5)

Publications & Preprints

Deciphering Transcriptome Complexity via Long-read Sequencing (Review)

Submitted to *iMeta*

Expected 2026 Wuhan, China

- **Chuwen Xu***, Co-first Author (Rank 1/4), et al.
- Developed a systematic computational taxonomy to categorize cutting-edge bioinformatics tools based on their underlying **algorithmic frameworks and data modeling principles** (e.g., error-correction hidden Markov models, splice-graph algorithms, and deep learning-based isoform clustering).
- Formulated reproducible statistical workflows and practical computational checklists to standardize isoform-level quantification, structural variation detection, and repeat-derived transcription analysis.

Research Experience

Research Assistant

Prof. Xiufen Zou's Group, Wuhan University

October 2025 – Present Wuhan, China

Under the supervision of Dr. Dingjie Wang

- **Project I: Algorithmic Landscapes & Roadmap for Long-read Transcriptomics** *(January 2026 – Present)*
  - Served as the **project lead and coordinator** for a systematic review and methodological roadmap on long-read sequencing and transcriptomics (submitted to *iMeta*).
  - Analyzed diverse bioinformatics algorithms by dissecting their underlying mathematical principles and connecting statistical foundations with multi-omics applications.
  - Drafted the technical manuscript and independently designed 5 conceptual figures and algorithmic workflows to illustrate complex computational models.

- **Project II: Multi-Platform Transcript Quantification Framework** *(May 2026 – Present)*
  - Contributed to the mathematical formulation and algorithmic design of a novel multi-platform transcript quantification framework.
  - Spearheaded large-scale bioinformatic data analysis, pipeline execution, and downstream statistical evaluation.
  - Developed statistical models based on linear algebra and probabilistic frameworks to characterize transcript and gene identifiability and quantify inference uncertainty.

Competitions

Mathematical Contest in Modeling (MCM)

COMAP (Honorable Mention)

Feb 2026

International

- Developed a multi-objective optimization framework integrated with the **Tsiolkovsky equation** and **NSGA-II** for lunar logistics layout.
- Modeled infrastructure queuing resilience via an **M/M/c finite-source model** and **Monte Carlo** simulations.

Central China Cup Math Modeling Contest

HBSIAM (3rd Prize ×2)

2025 & 2026

Regional

- **2026 (Bike Logistics):** Built demand models using **Cubic Spline Interpolation** and Dijkstra's algorithm; applied **Genetic Algorithms** to optimize dynamic routing.
- **2025 (Optical Mapping):** Formulated a cylinder reflection bijective mapping model using **Non-linear Programming** and **Differential Evolution** for image alignment.

National Mathematical Contest in Modeling

CUMCM Project Research

Sep 2025

National

- Formulated a 3D kinematic tracking and occlusion model for UAV tactical deployment against missile guidance systems.
- Implemented **Sequential Quadratic Programming (SQP)** and **Simulated Annealing** to maximize multi-agent dynamic defense time.